AFN-1069

BGE131

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

First Semester

Geology

GENERAL GEOLOGY – I

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.

Define the following :

1. Nebular hypothesis.

2. Ocean Basins.

3. Isostacy.

4. Tectonic mountains.

5. Volcanic bomb.

6. Mohorovicic discontinuity.

7. Geothermal gradient.

8. Asthenosphere.

9. Convergent boundaries.

10. Focus and Epicenter.

Section - B (5 × 7 = 35)

Answer **all** questions.

All questions carry equal marks.

Each answer should be of about 250 words.

11. (a) Discuss briefly about Relief features.

(*Or*)

(b) Brief about the physical properties of Earth like temperature, size, density etc.

12. (a) Explain about Airy's and Pratt's theories on Isostacy.

(*Or*)

(b) Discuss the classification of mountains.

13. (a) Describe the types of Volcanic Eruptions.

(*Or*)

(b) Brief about radioactive methods for dating rocks.

14. (a) Explain the Primary and Secondary waves.

(*Or*)

(b) Enumerate the concept of sea flow spreading.

15. (a) Briefly explain about the Solar system.

(*Or*)

(b) Explain about Ritcher scale for measuring Earthquake intensity.

Section - C (3 × 15 = 45)

Answer **all** questions.

All questions carry equal marks.

Each answer should be of about 1,200 words.

16. (a) Explain about various theories on the origin of the earth.

(*Or*)

(b) Explain about Mountain Building Orogeny.

17. (a) Explain about the interior of the Earth with a neat sketch.

(*Or*)

(b) Explain about the mid oceanic ridges.

18. (a) Explain about the concept of Plate tectonics.

(*Or*)

(b) Describe about the Earthquakes prone location in India and add a note on tectonics of earthquakes.

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AFN-1071 BGE231

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Second Semester

Geology

PALAEONTOLOGY AND GENERAL STRATIGRAPHY

(Non-CBCS 2004—onwards)

Duration : 3 Hours Maximum : 100 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Describe use of fossils.

2. Write shape of the shell in foraminifera.

3. Describe pentrimites.

4. Describe the fossil coral Favosites.

5. Sketch and label the features of pecten.

6. Describe the morphology of Nautilus.

7. Define Cephalon.

8. What is Glassopteris ?

9. Define Homotaxis ?

10. Describe biostratigraphic units.

Part - B (5 × 7 = 35)

Answer **all** questions.

11. (a) Write in brief about Fossils and paleoecology

Or

(b) Explain in brief about the following :

(i) Globigerina.

(ii) Fusulina.

12. (a) Distinguish between regular and irregular Echinoids.

Or

(b) Explain the morphological features of class Anthozoa (corals)

13. (a) Describe the types of suture lines in Cephalopoda.

Or

(b) Give a brief account on Brachial skeleton in Brachiopoda.

14. (a) Outline briefly the morphology of trilobita.

Or

(b) Write note on classification of graptolites.

15. (a) Describe the stratigraphic classification.

Or

(b) What is correlation ? Explain the types of correlation.

Part - C (3 × 15 = 45)

Answer **all** Questions.

16. (a) Describe the Geological distribution of foraminifera. Add a note on foraminifera in oil exploration.

Or

(b) Describe the morphology and classification of Echinoids.

17. (a) Explain the Morphology and Geological distribution of Lamellibranchia.

Or

(b) Explain the Morphology and shapes of Gastropoda shells.

18. (a) Write an essay on the Morphology and Geological distribution of Graptolites.

Or

(b) Describe the following fossils :

(i) Tetragraptus.

(ii) Diplograptus.

(iii) Sigillaria.

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AFN-1073 BGE331

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Third Semester

Geology

MINERALOGY

(Non-CBCS—2004 onwards)

Duration : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Define hardness.

2. Describe molecular and emphirical formula of minerals.

3. Explain Isotropism.

4. List any four special properties of Quartz.

5. Name any *two* Carbonate minerals with their chemical composition.

6. Give two examples for Biaxial minerals.

7. Describe single chain silicate structure with a sketch.

8. Describe Backe line test.

9. List the physical properties of Andalusite.

10. Write the optical characters of sphere.

Section - B (5 × 7 = 35)

Answer **all** questions.

11. (a) How are specific gravity of minerals determines ?

Or

(b) Explain Moh's scale of hardness.

12. (a) Write short notes on plane polarised light.

Or

(b) Give a brief account on Nicol prism.

13. (a) Explain in brief about Birefringence.

Or

(b) Briefly explain conoscopic characters of biaxial minerals.

14. (a) Bring out optical characters of Mica.

Or

(b) Give a short account of different crystalline quartz.

15. (a) Differentiate Pyroxenes from Amphiboles.

Or

(b) Describe the physical properties of Kayanite and Sillimanite.

Section - C (3 × 15 = 45)

Answer **all** questions.

16. (a) Explain the following :

(i) Isomorphism.

(ii) Polymorphism.

(iii) Pseudomorphism.

Or

(b) Elaborate the construction and uses of the following accessories.

(i) Quartz wedge.

(ii) Gypsum plate.

(iii) Mica plate

17. (a) Describe different optical properties of minerals with suitable examples.

Or

(b) Explain the following :

(i) Optic axial plane.

(ii) Uniaxial and biaxial interference figure.

18. (a) Write an essay on feldspar group of minerals.

Or

(b) Elaborate on the physical and optical properties, chemical composition and mode of coccurrences of Garnet group of minerals.

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AFN-1074 BGE431

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fourth Semester

Geology

INDIAN STRATIGRAPHY

(Non-CBCS—2004 onwards)

Duration : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.
Each answer should be in a word **or** a sentence.

1. Define Order of Superposition.

2. Give the rock types associated with Peninsular Gneisses.

3. Enumerate the classification of Triassic system.

4. Define Neobolus beds.

5. List out the division of lower Siwaliks.

6. Mention the division of Jurassic rocks found in Kashmir.

7. Give names of any three plant fossils.

8. Define Cenomanian Marine Transgression.

9. Define Bagh beds.

10. Lithological division of Jaintia series.

Section - B (5 × 7 = 35)

Answer **all** questions.

11. (a) Discuss the economic importance of Cudappah system in India.

Or

(b) Explain the major Physiographic divisions of India.

12. (a) Explain Cambrian of salt range and discuss the age of Saline series.

Or

(b) Discuss the Umaria Marine beds, its geographic location and environment of deposition.

13. (a) Explain the Jurassic system of Kutch.

Or

(b) Characteristics of Triassic spiti and their faunal assemblages.

14. (a) Discuss the age of Deccan traps.

Or

(b) Given an account of Inter trappean beds.

15. (a) Discuss the Kurnool system and add a note on the environment of depostion.

Or

(b) Describe the depositional environment and faunal assemblage of Siwaliks.

Section - C (3 × 15 = 45)

Answer **all** questions.
Each answer should be in about 1000 words.

16. (a) Compare and contrast on the drainage system of Peninsula and extra Peninsula.

Or

(b) Explain the distribution, characters and age of Dharwar super group in India.

17. (a) Describe the distribution and economic importance of Vindhyan system.

Or

(b) Give an account of Gondwana group of rocks found in India. Add a note on the division and condition of sedimentation.

18. (a) Write about the lemeta beds and its significance in Indian stratigraphy.

Or

(b) Explain the distribution, economic importance of Oligocene and Miocene rocks of Assam.

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AFN-1076 BGE532

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fifth Semester

Geology

IGNEOUS PETROLOGY

(Non-CBCS 2004—onwards)

Duration : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.
Each answer should be in a word **or** a sentence.

1. Describe the characteristics of an Igneous rock.

2. Give the composition of Magmas.

3. Explain Graphic texture with example.

4. What is Ropy lava ?

5. Describe C. I. P. W. Classification.

6. Explain Shand's saturation principle.

7. What is a gabbro ?

8. Mention any *two* ultrabasic rocks.

9. What is meant by eutectic.

10. Describe Assimilation.

Section - B (5 × 7 = 35)

Answer **all** questions.
Each answer should be in about 250 words.

11. (a) Give an account on the intensive forms of igneous rocks.

Or

(b) What are Pyroclastic deposits ? Explain them.

12. (a) Explain Amygdaloidal structure and its Petrological significance.

Or

(b) Describe Porphyritic texture with example.

13. (a) Give the principles in the classification of Igneous rocks.

Or

(b) Outline Rosenbusch and Johannsen's classification

14. (a) Describe the Petrographic character of Syenite.

Or

(b) What are alkaline rocks ? Give their mode of origin.

15. (a) Describe various theories of differentiation.

Or

(b) Explain incongruent melting crystallization with example.

Section - C (3 × 15 = 45)

Answer **all** questions.
All questions carry equal marks.
Each answer should be in about 1000 words.

16. (a) Describe and distinguish between an igneous, sedimentary and metamorphic rocks.

Or

(b) What is a magma ? Give their classification, constitution and composition.

17. (a) Describe with neat sketches the equigranular and inequigranular textures of igneous rocks.

Or

(b) Write about Tyrell's tabular classification.

18. (a) What are Anorthosites ? Give their Petrography and origin.

Or

(b) Describe Albite-Anorthite system with a diagram. Give their Petrological significance.

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AFN-1077 BGE533

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fifth Semester

Geology

SEDIMENTARY AND METAMORPHIC PETROLOGY

(Non-CBCS—2004 onwards)

Duration : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** the questions.

1. Explain about the decomposition of rocks.

2. Mention the four classified groups of sedimentary rocks.

3. What is meant by 'Rudaceous rocks' ?

4. What are 'Breccias' ?

5. What is understood by 'Guano' ?

6. What are 'Secretions' ?

7. Mention any *four* kinds of metamorphism.

8. Explain 'Cataclastic' structure.

9. What do you mean by 'Plutonic metamorphism' ?

10. Explain the term 'Metasomatism'.

Section - B (5 × 7 = 35)

Answer **all** the questions.
Each answer should be in about 250 words.

11. (a) Explain briefly any *one* of the sedimentary processes.

Or

(b) Explain any one non-elastic texture of sedimentary rocks.

12. (a) Briefly explain the important types of Soils.

Or

(b) Give the characteristic features of Conglomerates.

13. (a) Give a brief outline on Salt deposits.

Or

(b) Describe briefly the Siliceous sedimentary deposits of organic origin.

14. (a) Give a brief outline on the agents of metamorphism.

Or

(b) What are Migmatites ? How are they formed ?

15. (a) Give a brief account on 'Pneumatolytic' metamorphism.

Or

(b) Describe briefly the formation of Quartzite and gneiss.

Section - C (3 × 15 = 45)

Answer **all** the questions.
Each answer should be in about 1000 words.

16. (a) Give a broad outline on the classification of sedimentary rocks with examples.

Or

(b) Write an essay on the characteristic features and mode of formation of sedimentary residual deposits.

17. (a) Write an essay on the sedimentary organic deposits of carbonaceous origin.

Or

(b) Give a detailed account of the important metamorphic structures with neat sketches.

18. (a) Give an elaborate account of Plutonic metamorphism and its products.

Or

(b) Describe briefly the following :

(i) Siderite.

(ii) Palingenesis.

(iii) Charnockites.

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AFN-1078 BGE631

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Sixth Semester

Geology

ECONOMIC GEOLOGY

(Non-CBCS—2004 onwards)

Duration : 3 Hours Maximum : 100 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Define Sublimation.

2. Define Tenor.

3. Define Provinces.

4. Define Ore localisation.

5. List out the occurrences of iron ore in Tamil Nadu.

6. Write the uses of Zirconium.

7. Write the qualities required for the raw material in abrasives industry.

8. Write the uses of refractories.

9. Define Peat.

10. What are natural fuel and mention its uses.

Section - B (5 × 7 = 35)

Answer **all** questions.

11. (a) Explain Residual deposits with examples.

Or

(b) Give account on contact metasomatism.

12. (a) Give a account on geologic thermometers.

Or

(b) Explain the Lindgrens classification.

13. (a) Describe the origin and occurrences of copper ores in India.

Or

(b) Give an account Magnesium ores and their distribution in India.

14. (a) Give a short note on Ceramic and Cement Industry in India.

Or

(b) Explain the uses of abrasive minerals and their distribution in India.

15. (a) Describe the varieties of coal.

Or

(b) Explain the process of formation of petroleum.

Section - C (3 × 15 = 45)

Answer **all** questions.

16. (a) Give a detailed account on the origin of phosphatic deposits and metamorphism

Or

(b) Explain the Bateman classification of ore deposits.

17. (a) Write an essay on the mode of occurrences, uses and distribution of fertiliser and building stones in India.

Or

(b) Give a detailed account on metallogenic epochs and provinces of India.

18. (a) Explain the origin, occurrences and distribution of Petroleum in India.

Or

(b) Enumerate the Mineral wealth of Tamilnadu.

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AFN-1079 BGE632

B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Sixth Semester

Geology

APPLIED GEOLOGY

(Non-CBCS—2004 onwards)

Duration : 3 Hours Maximum : 100 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

Define the following :—

1. Photogeology.

2. Parallax.

3. Tenor.

4. Resistivity.

5. Darcy‘s law.

6. Magnetic method.

7. Crosscut.

8. Critical mineral.

9. Archdam.

10. Winze.

Part - B (5 × 7 = 35)

Answer **all** questions.
Each answer should be in about 250 words.

11. (a) Explain EMR at earth surface.

Or

(b) Explain briefly on fundamentals of air photo interpretation on landforms.

12. (a) Explain certain physiographic guides in geological prospecting.

Or

(b) Explain Geo-chemical prospecting of ore deposits.

13. (a) Outline the different types of aquifers.

Or

(b) Brief note on radioactive method of prospecting.

14. (a) Discuss the classification of mining methods.

Or

(b) Brief note on conservation of minerals.

15. (a) Outline geological conditions necessary for construction of dams and reservoirs.

Or

(b) Explain in details about fundamentals of air photo interpretation in water resources.

Part - C (3 × 15 = 45)

Answer **all** questions.
Each answer should be in about 1000 words.

16. (a) Explain mineralogical, lithologic and structural guides in geological prospecting.

Or

(b) Explain in detail about hydrologic cycle.

17. (a) Give account on electrical and seismic method of Geophysical prospecting.

Or

(b) Write in detail National Mineral policy.

18. (a) Outline geological conditions necessary for dams and reservoirs.

Or

(b) Enumerate the methods for prevention of coastal erosion.

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